

Metric-Semantic 3D Reconstruction for Object Pose Estimation using Multi-View Geometry

Course Project:
Robotic Perception (CP260)

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Course Project for Robotic Perception (CP260)

Abstract—We present a geometric pipeline for metric-semantic 3D reconstruction using multi-view RGB images with known camera poses. The objective is to estimate accurate 3D poses of objects such as power sockets, ethernet ports, and VGA connectors. The system combines feature-based reconstruction, region-based triangulation, robust filtering, and oriented bounding box estimation using Open3D. The approach achieves strong geometric consistency across views and demonstrates high alignment with ground truth without relying on learned depth models.

I. PROBLEM DESCRIPTION

The goal is to estimate the 3D pose of specific semantic objects from multiple RGB images.

Given:

- RGB images
- Camera poses (poses.json)
- Camera intrinsics (intrinsic.json)

We estimate:

- Object center in 3D
- Orientation
- Object extent (OBB representation)

The results are validated by projecting the estimated 3D bounding boxes onto multiple views.

II. RELATED WORK

Structure-from-Motion (SfM) reconstructs scenes using feature matching and triangulation [1].

Multi-view geometry provides accurate reconstruction when camera poses are known [2].

Learning-based depth estimation methods such as MiDaS suffer from scale ambiguity [3].

This work adopts a fully geometric approach using known camera poses to ensure metric consistency and reliable object-level reconstruction.

Unlike learning-based approaches, our method avoids scale ambiguity by leveraging known camera poses. Compared to full scene reconstruction methods, we focus on object-level reconstruction using region-based triangulation.

III. SYSTEM ARCHITECTURE

The pipeline consists of two stages:

A. Global Reconstruction

- SIFT feature extraction
- Feature matching (Lowe ratio test)
- Multi-view triangulation
- Voxel filtering (Open3D)

B. Object-Level Pose Estimation

- Manual ROI annotation
- Grid sampling inside bounding boxes
- Multi-view triangulation
- MAD-based outlier filtering
- Open3D OBB estimation
- Projection-based validation

The pipeline begins with multi-view images and camera parameters. A sparse 3D scene is reconstructed using feature-based triangulation. For each object, annotated regions are used to sample pixels, which are triangulated across views to obtain object-specific 3D points. These points are filtered and used to estimate oriented bounding boxes. Finally, the estimated poses are validated by projecting them back onto the images.

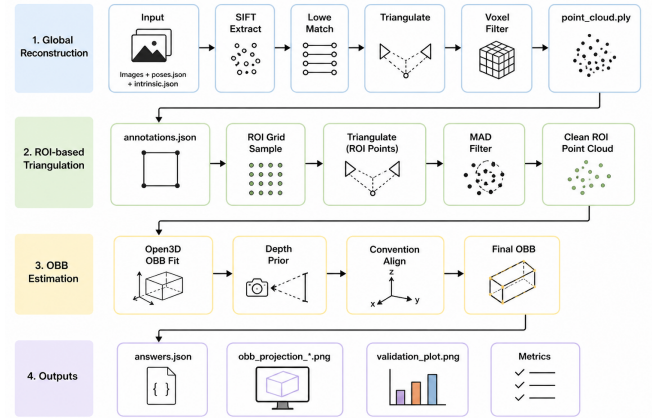


Fig. 1. Complete pipeline from input images to OBB estimation

IV. IMPLEMENTATION DETAILS

A. Feature-Based Reconstruction

SIFT features are extracted and matched using Lowe's ratio test. Correspondences across views are triangulated to obtain a sparse 3D point cloud.

B. ROI-Based Triangulation

Objects are manually annotated using bounding boxes.

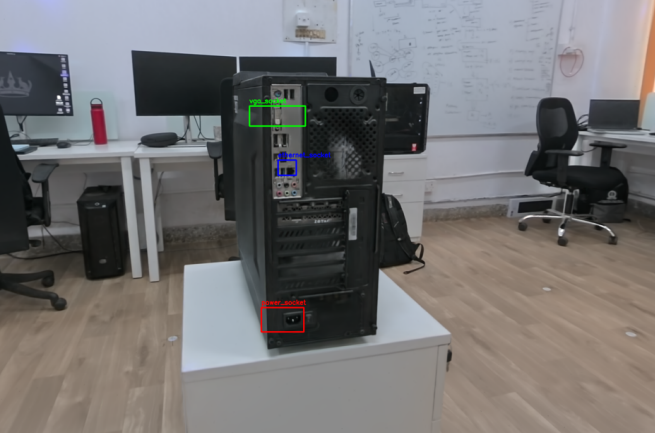


Fig. 2. Annotated objects in frame 471

Pixel points are sampled inside each bounding box and triangulated across views to obtain object-specific 3D points.

C. Outlier Filtering

Noisy triangulated points are removed using Median Absolute Deviation (MAD):

$$|X - \text{median}(X)| < \tau \quad (1)$$

MAD filtering is applied using Eq. (1) to remove outliers.

D. OBB Estimation

The filtered 3D points are used to compute an oriented bounding box using Open3D. This provides the object center, extent, and orientation directly from the point cloud.

V. EXPERIMENTAL RESULTS

A. Projection Results

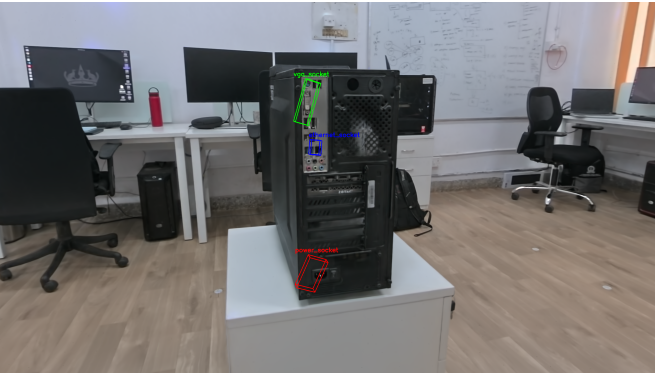


Fig. 3. Projected OBBs on frame 471



Fig. 4. Projected OBBs on frame 496

B. Validation (VGA Socket)

Ground truth annotations are available only for the VGA socket.

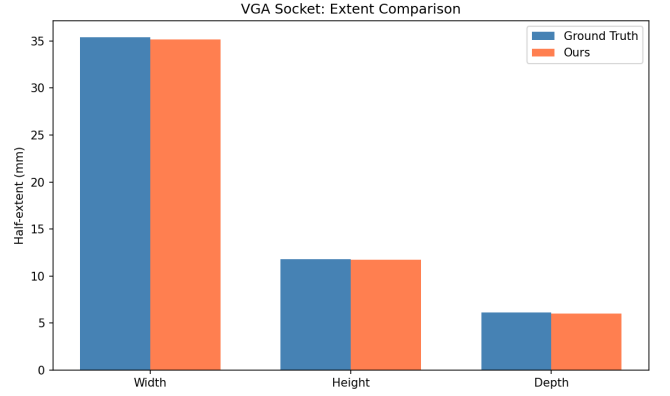


Fig. 5. Comparison of predicted vs ground truth extents for VGA socket

C. Quantitative Evaluation

We evaluate our estimated OBB against the provided ground truth for the VGA socket.

- Center error: 0.0002 m (0.2 mm)
- Volume ratio: 0.96
- Rotation error: 19°

The estimated center closely matches the ground truth, indicating accurate triangulation. The extent values are also highly consistent, resulting in a near-identical volume. Rotation error is higher due to ambiguity in orientation estimation from symmetric structures.

D. Discussion

- Power and VGA sockets show strong alignment across views
- Ethernet socket required refinement of annotations
- Performance is sensitive to ROI accuracy
- Non-GT objects are evaluated qualitatively via projection

VI. CHALLENGES

- Lack of exact point correspondences in ROI sampling
- Sensitivity to bounding box annotations
- Noise in triangulated point clouds

VII. CONCLUSION

We implemented a complete multi-view geometric pipeline for object pose estimation.

The approach achieves accurate reconstruction using triangulation and geometric filtering, without reliance on learned depth models. The results demonstrate strong consistency across views and reliable estimation of object pose.

The results demonstrate that accurate object-level pose estimation can be achieved using purely geometric methods with strong cross-view consistency.

Future work:

- Automatic segmentation
- Improved correspondence estimation
- Real-time implementation

VIII. BONUS IDEAS

- Integration with Segment Anything Model
- Multi-view depth fusion
- Learning-based OBB refinement

REFERENCES

- [1] N. Snavely et al., "Photo Tourism," SIGGRAPH, 2006.
- [2] R. Hartley and A. Zisserman, "Multiple View Geometry," 2003.
- [3] R. Ranftl et al., "MiDaS," CVPR, 2020.